

4 Measures

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM m	Use and interpret maps and scale drawings	• Use and interpret maps and scale drawings • Read and construct scale drawings • Draw lines and shapes to scale • Estimate length using a scale diagram
GM n	Understand the effect of enlargement for perimeter, area and volume of shapes and solids	• Understand the effect of enlargement for perimeter, area and volume of shapes and solids • Understand that enlargement does not have the same effect on area and volume • Use simple examples of the relationship between enlargement and areas and volumes of simple shapes and solids
GM o	Interpret scales on a range of measuring instruments and recognise the inaccuracy of measurements	• Indicate given values on a scale • Interpret scales on a range of measuring instruments – seconds, minutes, hours, days, weeks, months and years – mm, cm, m, km, ml, cl, l, mg, g, kg, tonnes, °C • Use correct notation for time, 12- and 24-hour clock • Work out time intervals • Know that measurements using real numbers depend upon the choice of unit • Recognise that measurements given to the nearest whole unit may be inaccurate by up to one half in either direction

Ref	Content descriptor	Concepts and skills												
GM p	Convert measurements from one unit to another	- Convert between units of measure within one system - Convert metric units to metric units (Metric equivalents should be known) - Convert imperial units to imperial units (NB: Conversion between imperial units will be given) - Know rough metric equivalents of pounds, feet, miles, pints and gallons <table><thead><tr><th>Metric</th><th>Imperial</th></tr></thead><tbody><tr><td>1 kg</td><td>2.2 pounds</td></tr><tr><td>1 l</td><td>$1\frac{3}{4}$ pints</td></tr><tr><td>4.5 l</td><td>1 gallon</td></tr><tr><td>8 km</td><td>5 miles</td></tr><tr><td>30 cm</td><td>1 foot</td></tr></tbody></table> - Convert between metric and imperial measures - Estimate conversions - Convert between metric area measures - Convert between metric volume measures - Convert between metric speed measures - Convert between metric measures of volume and capacity eg $1\text{ cm}^3 = 1\text{ ml}$	Metric	Imperial	1 kg	2.2 pounds	1 l	$1\frac{3}{4}$ pints	4.5 l	1 gallon	8 km	5 miles	30 cm	1 foot
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GM q	Make sensible estimates of a range of measures	- Make sensible estimates of a range of measures in everyday settings - Choose appropriate units for estimating or carrying out measurement												
GM r	Understand and use bearings	- Use three figure-bearings to specify direction - Mark on a diagram the position of point B given its bearing from the point A - Give a bearing between the points on a map or scaled plan - Given the bearing of point A from point B, work out the bearing of B from A												

Foundation

Ref	Content descriptor	Concepts and skills
GM s	Understand and use compound measures	Understand and use compound measures including speed
GM t	Measure and draw lines and angles	- Measure and draw lines, to the nearest mm - Measure and draw angles, to the nearest degree
GM u	Draw triangles and other 2-D shapes using ruler and protractor	- Make accurate drawing of triangles and other 2-D shapes using a ruler and a protractor - Make an accurate scale drawing from a diagram - Use accurate drawing to solve bearings problems